Literature Review: Credit Union Balance Sheet Forecasting

Combined edition · November 2026

This file combines the five literature reviews behind the forecasting workbooks into one document. Each chapter covers the economic channels, the macro series that measure them, the expected sign on each target, the published models used in the workbooks, and the references.

Suggested run order: loan and deposit balances (chapters 1–2) and rates (chapter 4) first; then allowances (chapter 3); then income, losses, RWA and capital (chapter 5), which take the earlier forecasts as inputs.

The mortgage loan review is not included here; it was written in an earlier session and can be added as a sixth chapter.

#	Chapter	Page
1	Commercial and personal loan balances	2
2	Deposits (demand, term) and investments	17
3	Loan loss allowances (mortgage, commercial, personal)	29
4	Loan and deposit interest rates	41
5	Income, losses, RWA and Tier 1 / Tier 2 capital	54

Literature Review: Commercial and Personal Loan Balances

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3. Literature review

This review applies the structure of the mortgage review to the two other Ontario loan targets: commercial (business) loans and personal (consumer) loans. It covers five strands, plus recent work:

- Theory of why business and household credit move with the economy.
- Empirical credit-demand models for each loan type.
- Canadian evidence, with emphasis on the Bank of Canada.
- How central banks and supervisors project balance sheets in stress tests.
- Methods for forecasting with many predictors.
- Recent developments, 2019–2026: credit-line drawdowns and the COVID shock, dynamic balance sheets in stress tests, new Canadian data on cards and HELOCs, machine-learning forecasting, and model-risk regulation.

The foundational studies (Sections 3.1–3.5) still define the standard specification for each loan type. Section 3.6 reviews the recent work that updates it, Section 3.7 summarises the models, and Section 3.8 draws out the design implications. Sections 3.4 and 3.5 apply to all three loan targets and are kept short where they repeat the mortgage review.

3.1 Why credit moves with the cycle: theory

As for mortgages, loan volumes are an equilibrium of credit demand and credit supply. The two loan types differ in what drives demand and in how the cycle shows up in balances.

Commercial loans. Firms borrow to finance working capital (inventories, receivables, payroll) and fixed investment, so demand rises with sales, activity and investment and falls with the cost of credit. On the supply side, the bank lending channel (Bernanke & Blinder, 1992; Kashyap & Stein, 2000) and the broader credit channel (Bernanke & Gertler, 1995) imply that policy rates and bank funding conditions shift loan supply. The financial accelerator (Bernanke, Gertler & Gilchrist, 1999) and collateral models (Kiyotaki & Moore,

1997) explain why firm net worth, equity prices and commercial property values amplify the cycle: weaker balance sheets raise the external finance premium and tighten borrowing constraints.

Two features make business loans less pro-cyclical in the short run than household loans:

- After a monetary tightening, business loans often rise for several quarters while household loans fall, because firms finance inventories and lost cash flow (den Haan, Sumner & Yamashiro, 2007).
- Many business loans are drawn on committed credit lines. In a downturn, firms draw on those lines, so balances can rise while new lending falls (Ivashina & Scharfstein, 2010).

Personal loans. Households borrow on credit cards, unsecured lines of credit, auto loans and (in Canada) home equity lines of credit (HELOCs) to smooth consumption against income and to buy durables. In permanent-income models with liquidity constraints, borrowing rises with expected income and falls with the interest rate, and credit growth predicts consumption when constraints bind (Ludvigson, 1999; Bacchetta & Gerlach, 1997). Housing collateral links personal credit to house prices: rising home equity relaxes borrowing limits and is extracted through HELOCs (Mian & Sufi, 2011; Aron et al., 2012). Unemployment raises default risk, which leads lenders to tighten and write off balances (Gross & Souleles, 2002b).

These mechanisms give the expected signs used in the two workbooks:

- Activity, investment and business confidence: positive for commercial loans.
- Income, consumption and consumer confidence: positive for personal loans.
- House prices: positive for personal loans (HELOC collateral); commercial property and construction positive for commercial loans.
- Loan interest rate (prime): negative for both; the policy rate has no sign for commercial loans because its short-run effect changes sign with the horizon.
- Unemployment, insolvencies, delinquencies and credit spreads: negative.

3.2 Empirical credit-demand models

Commercial loans. Aggregate credit-demand studies use the same long-run form as for mortgages. Calza, Manrique & Sousa (2006) model euro-area private-sector loans with a cointegrated VAR: real loans rise with real GDP and fall with a composite bank lending rate. Hofmann (2004) finds that credit cointegrates with GDP and the real rate only once property prices are added, and Davis & Zhu (2011) show that commercial property prices

and bank lending move together across countries. Brigden, Chrystal & Mizen (2000) estimate separate UK equations for companies and households, and find that corporate borrowing responds more to investment and less to income than household borrowing.

For the short run, lending-survey and credit-spread studies add the supply side:

- Lown & Morgan (2006) find that the US loan officer survey's lending standards predict business loans and output better than interest rates do.
- Bassett, Chosak, Driscoll & Zakrajšek (2014) build a lending-standards index purged of demand factors; a tightening lowers business lending capacity and output.
- Gilchrist & Zakrajšek (2012) show that the excess bond premium, the part of corporate spreads not explained by default risk, predicts investment and activity.
- Becker & Ivashina (2014) and Adrian, Colla & Shin (2013) show that when bank supply tightens, firms with bond access issue bonds instead: in 2007–09 bank lending to firms fell while bond financing rose. Kashyap, Stein & Wilcox (1993) find a similar shift away from bank loans after monetary tightening.

Personal loans. Ludvigson (1999) finds that consumer credit growth is predicted by expected income and is itself a predictor of consumption, and Bacchetta & Gerlach (1997) find the same across five countries. Fernández-Corugedo & Muellbauer (2006) build a UK credit-conditions index from unsecured and secured debt equations. Its key insight is that the supply side shifts the whole relationship over time. Micro studies give the elasticities behind the expected signs:

- Gross & Souleles (2002a): higher card limits raise debt immediately, and balances respond to interest rates.
- Attanasio, Goldberg & Kyriazidou (2008): car-loan demand responds more to loan maturity than to the interest rate for younger and lower-income borrowers.
- Carroll, Fuhrer & Wilcox (1994): consumer sentiment forecasts household spending beyond income.
- Agarwal, Chomsisengphet, Mahoney & Stroebel (2018): banks pass credit expansions through least to the borrowers who most want to borrow, so supply limits matter most for riskier households.
- Dynan (2012) and Mian, Rao & Sufi (2013): highly leveraged households cut spending and borrowing more after a shock, so high debt-to-income lowers future credit growth.

3.3 Canadian evidence: Bank of Canada and others

Commercial loans.

- Hogg (2021, BoC SAN 2021-13) uses microdata from the Bank's Senior Loan Officer Survey (SLOS). A tightening in supply lowers business-loan growth for up to nine quarters, while the demand signal fades after about two quarters. In the 2008–09 crisis, supply contributed about −5 percentage points and demand about −1.5 points to business-loan growth.
- Duttagupta & Barrera (2010, IMF WP 10/5) use a Bayesian VAR and find that financial conditions, including lending standards, explain about one-third of the variability of Canadian real GDP growth, so credit and activity are jointly determined.
- Champagne & Gouin-Bonenfant (2022, BoC SWP 2022-49) build a firm-level measure of credit constraints from survey data on whether financing requests were granted. Credit constraints account for roughly a third of the SME employment response to monetary policy, which is consistent with a strong financial accelerator.
- Sekkel, Takamura & Terajima (2024, BoC SWP 2024-41) use loan-level data to trace how policy-rate changes reach the performance of small business loans.

Personal loans. The Canadian market differs from the US because HELOCs are a large share of non-mortgage household credit:

- The Financial Consumer Agency of Canada (2017) reports that HELOC balances grew from $35 billion in 2000 to $211 billion in 2016, reaching nearly 40% of non-mortgage consumer debt by 2010. About 80% of HELOCs are sold within readvanceable mortgages.
- Bailliu, Kartashova & Meh (2012, Bank of Canada Review) attribute the rise in household borrowing to rising house prices, low interest rates and financial innovation, especially home-equity extraction.
- OSFI's Guideline B-20 (June 2012) capped the revolving HELOC portion at 65% of the home's value, a regulatory break in personal-credit growth.
- Al-Mqbali, Bilyk, Caputo & Younker (2019, BoC SAN 2019-14) use new regulatory data to show that growth in combined mortgage-HELOC plans came from the amortizing mortgage part, and that HELOC balances fell as rates rose and consumption slowed. HELOCs therefore respond to interest rates and spending.
- Muellbauer, St-Amant & Williams (2015, BoC SWP 2015-40) show that a latent credit-conditions index matters for Canadian household debt and consumption, the same supply shift that Fernández-Corugedo & Muellbauer (2006) find in the UK.

3.4 Stress-testing practice: how balance sheets are projected

The practices summarised in the mortgage review apply to business and consumer portfolios too:

- The Federal Reserve's 2025 supervisory methodology assumes firms keep their current level of assets, including loans, over the nine-quarter horizon. The IMF's 2025 FSAP stress test for Canada also uses a static balance sheet.
- The NY Fed CLASS model (Hirtle, Kovner, Vickery & Bhanot, 2016) assumes fixed asset growth and projects income and loss ratios with an AR(1) plus macro variables, including the unemployment rate, real GDP, house prices, Treasury and BBB yields, spreads and equity prices.
- The Bank of Canada's MFRAF (Fique, 2017) covers solvency and liquidity feedback for systemically important banks, and HRAM (Peterson & Roberts, 2016) stress-tests household balance sheets at the micro level, which is the natural home for consumer-credit stress.
- Hipp & Ojea Ferreiro (2026, BoC SWP 2026-26) find that lending is stable while capital buffers are comfortable and contracts sharply near regulatory thresholds.

Two approaches are adopted directly, as for mortgages: the quantile autoregressions of Covas, Rump & Zakrajšek (2014), and the lag-expand, LASSO-select and summarise pipeline of Kapinos & Mitnik (2016).

One point is specific to commercial loans. Because committed lines are drawn in a downturn (Ivashina & Scharfstein, 2010; Greenwald, Krainer & Paul, 2025), a severe scenario can raise business-loan balances in the first quarters before they fall. A demand model with lagged drivers will not reproduce that jump, so it needs an overlay in severe scenarios.

3.5 Forecasting with many predictors

The methods are the same as for mortgages:

- Factor models built from large panels or from targeted predictors (Stock & Watson, 2002; Bai & Ng, 2008).
- Shrinkage with LASSO, ridge and the Elastic Net (Tibshirani, 1996; Zou & Hastie, 2005).
- Equal-weight forecast combinations, which are hard to beat (Bates & Granger, 1969; Stock & Watson, 2004).

- Random forests and gradient boosting for non-linearity (Breiman, 2001; Friedman, 2001). On small macro samples, statistical models often match or beat them (Makridakis, Spiliotis & Assimakopoulos, 2018).

For consumer credit, machine learning is most useful at the account level. Butaru et al. (2016) predict credit-card delinquency with machine learning on account, bureau and macro data from six large US banks, and find that risk drivers and their sensitivities differ widely across banks. This supports using account-level models for card losses and keeping the aggregate balance model small.

3.6 Recent developments (2019–2026)

Credit lines and the COVID shock (commercial).

- Greenwald, Krainer & Paul (2025, Journal of Finance) find that US bank lending to firms rose after COVID-19 began because large firms drew on their credit lines. Banks with larger drawdowns cut term lending, which squeezed smaller firms. Credit lines raise total bank credit in bad times but move it towards firms that invest less.
- Chodorow-Reich, Darmouni, Luck & Plosser (2022, Journal of Financial Economics) show that small firms have shorter, more collateralised credit lines and did not draw them in 2020 even when hit as hard as large firms. Aggregate business-loan balances in a downturn are therefore driven by large firms.

The COVID quarters should therefore stay excluded or dummied for commercial loans, as for mortgages: the balance spike reflects drawdowns, not demand from the macro drivers.

Canadian household credit, 2020–2026 (personal).

- Statistics Canada (2021) reports that non-mortgage household debt fell by a record $20.6 billion between February 2020 and January 2021, mostly because credit-card balances dropped 18.3% (from $90.6 billion to $74 billion). Income support and limited spending drove the fall, so COVID is a break for personal loans as well.
- Xiao (2024, BoC SAN 2024-18) uses TransUnion data on nearly 58 million card accounts. Carrying a balance, and using more than 80% of the limit, sharply raises the chance of missing other debt payments. Card utilisation is an early-warning indicator of household stress.
- Al Aboud, Sheikh, Su & Xu (2025, BoC SAN 2025-1) describe the Bank's enhanced dataset on outstanding mortgages and HELOCs at federally regulated lenders, a source for separating HELOCs from other personal credit.

Dynamic balance sheets in stress testing. The ECB's BEAST (Budnik et al., 2023) projects loan volumes dynamically by splitting loan growth into demand and supply. Demand depends on real GDP growth, inflation, the change in unemployment, the short-term rate, the term spread, the lending rate and lagged loan growth. Supply depends on bank capital, non-performing loans, profitability and funding costs. The same split applies here: the macro models are demand-side, and supply constraints in severe scenarios need a separate overlay.

Big data and machine learning. Giannone, Lenza & Primiceri (2021) warn against relying on one sparse model, Goulet Coulombe et al. (2022) find that non-linearity is the main gain from machine learning in periods of stress, and Bluwstein et al. (2023) find credit growth and the yield-curve slope to be the main predictors of crises. Fortin-Gagnon et al. (2022) provide a public Canadian and provincial database for cross-checking the Moody's factors.

Model-risk regulation. OSFI's revised Guideline E-23 on Model Risk Management (published September 2025, effective May 2027) applies to these models as to the mortgage model.

3.7 Summary of the reviewed models

Study	Loan type	Dependent variable	Main drivers	Method	Used here as
Calza, Manrique & Sousa (2006), ECB	Both	Real private-sector loans	Real GDP (+), lending rate (−)	Cointegrated VAR / VECM	ECM long run (both workbooks)
Hofmann (2004); Davis & Zhu (2011), BIS	Both	Real private credit; bank lending	GDP (+), property prices (+), real rate (−)	Cointegrated VAR; panel	Property and construction drivers
Brigden, Chrystal & Mizen (2000), BoE	Both	Corporate and household borrowing	Investment, income, rates	Sector equations	Separate driver sets by loan type
den Haan, Sumner & Yamashiro (2007)	Both	Loan categories after a policy shock	Policy rate	VAR	No sign on policy rate (commercial)

Study	Loan type	Dependent variable	Main drivers	Method	Used here as
Lown & Morgan (2006); Bassett et al. (2014), Fed	Commercial	Business loans, output	Lending standards (− when tighter)	VAR	Lending-survey model
Gilchrist & Zakrajšek (2012)	Commercial	Investment, activity	Credit spread (−)	Regression, VAR	Credit-channel model
Hogg (2021), BoC	Commercial	Business-loan growth	SLOS supply and demand	Panel regression	Lending-survey model, lag length
Ivashina & Scharfstein (2010); Greenwald, Krainer & Paul (2025)	Commercial	Bank lending in crises	Credit-line drawdowns	Loan-level	COVID exclusion, severe-scenario overlay
Ludvigson (1999); Bacchetta & Gerlach (1997)	Personal	Consumer credit, consumption	Income (+), rates (−)	Time-series regression	Consumer credit and spending model
Fernández-Corugedo & Muellbauer (2006); Muellbauer, St-Amant & Williams (2015)	Personal	Unsecured and secured debt	Credit conditions, income, rates, wealth	ECM system	Unsecured lending model, ECM
Mian & Sufi (2011); FCAC (2017); Al-Mqbali et al. (2019)	Personal	Home-equity borrowing	House prices (+), rates (−)	Micro data; regulatory data	Housing collateral (home equity) model
Gross & Souleles (2002a, 2002b)	Personal	Card debt; delinquency	Credit limits, rates, unemployment	Account-level panel	Signs on rates, unemployment, delinquency
Xiao (2024), BoC; Butaru et al. (2016)	Personal	Card delinquency	Utilisation, balances, macro	Panel; machine learning	Delinquency driver; account-level losses

Study	Loan type	Dependent variable	Main drivers	Method	Used here as
Hirtle et al. (2016), NY Fed CLASS	Both	Income and loss ratios	Unemployment, GDP, house prices, yields, spreads, equities	AR(1) + macro	Benchmark form (AR + macro)
Budnik et al. (2023), ECB BEAST	Both	Loan volumes (demand and supply)	GDP, inflation, Δunemployment, rates; bank capital	Semi-structural panel	Demand-side framing

3.8 Implications for these models

1. **Model the growth rate with persistence, as for mortgages.** Use the AR + macro form, with organic growth as the dependent variable.
2. **Use different core drivers for each loan type.** Commercial loans: real GDP, business investment, the prime rate and the credit spread. Personal loans: disposable income, house prices, the prime rate and unemployment. These are the "four-channel core" models in the two workbooks.
3. **Expect shorter adjustment than for mortgages.** Business lines and consumer credit reprice with prime and have no five-year renewal cycle, so use rate lags of one to two quarters rather than the long mortgage lags.
4. **Leave no sign on drivers whose effect changes with the horizon.** The policy rate and bond yields can raise business loans in the short run (den Haan et al., 2007; Becker & Ivashina, 2014), and the saving rate and financial wealth are ambiguous for personal loans.
5. **Add the lending survey when it is mapped.** The SLOS supply signal leads business-loan growth by up to nine quarters (Hogg, 2021), so mapping it is the most valuable data addition for commercial loans.
6. **Control for breaks.** COVID is a break for both loan types (credit-line drawdowns in business loans; card paydowns in personal loans). B-20 in 2012 is a break for personal loans if the series includes HELOCs.
7. **Benchmark against factor models and combinations, and do not trust one sparse model** (Giannone et al., 2021).
8. **Treat the models as demand-side.** Add an overlay in severe scenarios for credit-line drawdowns (commercial) and for lender tightening and charge-offs (personal), following BEAST (2023) and Hipp & Ojea Ferreiro (2026).

Appendix: using the workbooks

Each loan type has a literature workbook and a metadata file; point a config at both and run as usual.

File	What it holds
`config/commercial_loans_literature_workbook.xlsx`	15 concepts, 4 published models (core: four-channel business core), error-correction and regime settings, 19 references
`config/personal_loans_literature_workbook.xlsx`	14 concepts, 4 published models (core: four-channel consumer core), error-correction and regime settings, 19 references
`data/macro_metadata_commercial_loans.csv`	The 400 macro series with signs for business loan growth; `sign_reason` and `changed_from_mortgage` explain every sign
`data/macro_metadata_personal_loans.csv`	The same for personal loan growth

The config lines for a commercial run (personal is the same with `personal`):

```
data:
  metadata:
    file: data/macro_metadata_commercial_loans.csv
literature:
  workbook: config/commercial_loans_literature_workbook.xlsx
```

Both workbooks pass the pre-flight check and run end to end on the example data, so the specifications, signs and error-correction settings all work. The example's target is a mortgage balance, so the estimates themselves say nothing about these loan types yet.

Before a real run:

- [ ] Map a Senior Loan Officer Survey series to `credit_conditions` in both workbooks (turned so that easing is positive).
- [ ] Check which products the personal balance includes (HELOCs, cards, auto) and switch concepts off with `include = no` where needed.

- [ ] Decide how to treat the 2012 B-20 break for personal loans: a step dummy or a later start date.

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Scholar links open a Google Scholar search for the paper; the others go to the paper, its working-paper version or the publisher's page. References marked † are carried over from the mortgage review and were not re-checked here.

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Literature Review: Deposits and Investments

Nov 5, 2026 · @Xiaokai

Scope and approach

Demand deposits follow transactions and the opportunity cost of holding cash; term deposits follow the gap between term and demand rates; investments are mostly the balancing item between deposits and loans. Each balance gets the same treatment as the loan reviews: the economic channels, the series that measures each, the expected sign on quarterly growth, and the studies behind it.

- **Demand deposits**: chequing and savings accounts payable on demand, including high-interest savings. These pay little or nothing and are used for payments.
- **Term deposits**: fixed-term deposits and GICs (guaranteed investment certificates), including those held in registered plans. They pay a rate fixed for the term and are not used for payments.
- **Investments**: the credit union's securities and liquidity holdings: deposits with a central or other financial institution, government and other bonds, and other liquid assets held to meet liquidity rules.

As for loans, the target is quarterly log growth of the balance, so every sign below is a sign on growth. Series names in the tables follow the loan reviews; where the macro data has no matching series yet, the table says what to map.

Three points shape this review:

- **Rates matter through spreads, not levels.** What moves money between demand deposits, term deposits and market assets is the gap between their rates. A higher policy rate widens the gap between market rates and demand-deposit rates, so demand deposits flow out and term deposits flow in.
- **Deposits are partly created by lending.** A new loan credits the borrower's deposit account, so deposit growth and loan growth move together at the system level (McLeay, Radia & Thomas 2014). Loan growth is therefore a driver to watch, but it is not exogenous.
- **Sign 0 means no prior, not no effect.** As in the loan reviews, it marks drivers whose sign the literature finds changes with the horizon or the product.

Demand deposits: drivers and expected signs

Demand deposits behave like money: they grow with nominal income and fall when market rates rise, because the rate they pay lags far behind. Long-run money demand is stable once measured properly, with a strong negative link to short rates (Lucas & Nicolini 2015; Benati et al. 2016). When the policy rate rises, banks widen deposit spreads and deposits flow out, most strongly for chequing and savings accounts (Drechsler, Savov & Schnabl 2017).

Channel	Driver (series)	Sign	Why	Evidence
Transactions demand	Disposable income (FYPD.ION); real GDP (FGDPR.ION)	+	More spending and income means larger payment balances	Lucas & Nicolini (2015); Benati et al. (2016)
Nominal scale	Inflation (FCPI.ION)	+	Balances are nominal and scale with prices	Lucas & Nicolini (2015)
Opportunity cost	Overnight rate (FRBOC.IC)	−	Demand-deposit rates lag market rates, so higher rates push money into term deposits and market assets	Drechsler, Savov & Schnabl (2017); Hannan & Berger (1991)
Substitution to term	Term-deposit rate minus demand-deposit rate (map the posted rates)	−	A wider gap pays depositors to lock money up	Drechsler, Savov & Schnabl (2017)
Flight to safety	Market volatility (FVIXC.IC); corporate spread (FRSPREAD.IC)	+	In market stress, money moves into insured bank deposits	Gatev & Strahan (2006)
Saving	Household saving rate (FSAVR.IC)	+	Unspent income lands first in chequing and savings accounts	StatCan national balance sheet accounts

Channel	Driver (series)	Sign	Why	Evidence
Lending creates deposits	Household credit (FCRHH.IC); business credit (FCRBUS.IC)	+	New loans are paid into deposit accounts	McLeay, Radia & Thomas (2014)
Wealth versus portfolio shift	Equity prices (FSTSX.IC)	0	Wealth raises balances, but strong returns pull money into investments	Lucas & Nicolini (2015)
Labour market	Unemployment (FLBR.ION)	0	Lower income cuts balances, but precautionary saving raises them	Gatev & Strahan (2006)

The rate effect is about the gap between market rates and the deposit rate, not the rate level itself. Demand-deposit rates adjust slowly and only partly, and more slowly upward than downward (Hannan & Berger 1991; Neumark & Sharpe 1992), so a rise in the overnight rate widens the gap and drains balances for several quarters.

Term deposits: drivers and expected signs

Term deposits move the other way from demand deposits when rates change: a higher policy rate raises GIC rates much more than chequing rates, so money moves into term deposits. In the U.S., deposit spreads widen far less for time deposits than for chequing and savings accounts (Drechsler, Savov & Schnabl 2017). In Canada, households moved money from chequing and savings into fixed-term deposits through the 2022–23 hikes, and fixed-term deposits kept growing while other deposits fell (Statistics Canada, 13 September 2023).

Channel	Driver (series)	Sign	Why	Evidence
Rate gap	Overnight rate (FRBOC.IC), change	+	GIC rates follow market rates closely; demand rates do not, so the gap opens as rates rise	Drechsler, Savov & Schnabl (2017); Statistics Canada (2023)

Channel	Driver (series)	Sign	Why	Evidence
Rate gap, direct	Term-deposit rate minus demand-deposit rate (map the posted rates)	+	The return for locking money up	Drechsler, Savov & Schnabl (2017)
Income and wealth	Disposable income (FYPD.ION)	+	More saving to place	Lucas & Nicolini (2015)
Saving	Household saving rate (FSAVR.IC)	+	Saved income is partly locked into GICs	Statistics Canada national balance sheet accounts
Competing assets	Equity prices (FSTSX.IC)	–	Strong markets pull savings into funds and equities	Drechsler, Savov & Schnabl (2017) (households substitute into bonds and other assets)
Safety	Market volatility (FVIXC.IC)	+	Insured, fixed-rate deposits gain in market stress	Gatev & Strahan (2006)
Lender's funding need	Mortgage credit (FCRMORT.IC)	+	Lenders raise GIC rates to fund loan growth; credit unions fund mostly with deposits	McLeay, Radia & Thomas (2014)
Ageing	Population 65+ (Demographics theme)	+	Older households hold more of their savings in GICs	Plausible but not tested in the studies here; leave out if there is no series
Yield curve	Term spread (10-year minus overnight)	0	A steep curve favours longer GICs over short ones, but also favours bonds over GICs	No settled evidence

Term deposits react with a lag set by their maturity. A five-year GIC booked in 2023 cannot move until 2028, so balances respond to rate changes through new bookings and maturities, not the stock. When rates fall, maturing GICs tend to roll into demand deposits rather than into equities (Desjardins 2025), which turns the 2022–23 shift around.

Deposits: where the literature disagrees, and Canadian specifics

The policy rate moves total deposits less than it moves the split between demand and term, so the two balances need separate models with opposite rate signs.

- **Total outflow or reshuffle?** In the U.S., a rate rise drains total deposits into bonds and money funds; a 100 bp rise widens spreads by about 61 bp on average, and a typical 400 bp hiking cycle implies deposit outflows of about $1.2 trillion (Drechsler, Savov & Schnabl 2017). The effect is larger where the lender has more market power. Credit unions price GICs competitively to hold members, so more of the rate effect shows up as a shift into term deposits than as an outflow.
- **Stress brings money in, unless the stress is at your institution.** In market-wide stress, money flows into insured banks (Gatev & Strahan 2006). When doubt falls on one institution, uninsured and large depositors leave first (Iyer & Puri 2012), and deposit pricing and fragility feed on each other (Egan, Hortaçsu & Matvos 2017). Macro stress variables therefore carry a + sign; an institution's own risk is a separate, negative effect that macro drivers cannot capture.
- **Product changes break the series.** Money demand looked unstable in the 1980s until new deposit types were counted (Lucas & Nicolini 2015). The same risk applies to high-interest savings accounts: decide whether they sit in demand deposits and keep the definition fixed over the sample.

Canadian and credit union specifics:

- **Deposit insurance.** FSRA's Deposit Insurance Reserve Fund covers insurable deposits in non-registered accounts at Ontario credit unions up to $250,000 per depositor, principal and interest combined, and covers insurable deposits in registered accounts (RRSPs, TFSAs and others) without limit (FSRA). Registered GICs are therefore fully insured, which makes term deposits a very stable funding source.
- **The pandemic surge.** Chequing deposits at the chartered banks rose by nearly $200 billion from February 2020 to summer 2022, then fell as about $125 billion flowed into fixed-term deposits once hikes began in March 2022 (Globe and Mail). Both balances need a 2020–21 dummy or an estimation window that handles it.

- **The mix follows the rate level.** Term deposits used to exceed demand deposits; after rates hit their floor in the global financial crisis, the two swapped places (C.D. Howe Institute). A long sample therefore spans two regimes, which supports error-correction or regime models with the rate level as the threshold.

Deposits: published models for the workbook

Three fixed specifications and one error-correction model per balance, written in the same form as the loan workbooks; the four-channel cores feed the hybrid, regime and network models.

Demand deposits

Model	Drivers (form, lag)	Source
Money demand	Disposable income (q/q, L0) +; overnight rate (change, L1) −; inflation (q/q, L0) +	Lucas & Nicolini (2015); Benati et al. (2016)
Deposits channel	Real GDP (q/q, L1) +; overnight rate (change, L0) −; overnight rate (change, L1) −	Drechsler, Savov & Schnabl (2017)
Flight to safety	Market volatility (change, L0) +; corporate spread (change, L0) +; disposable income (q/q, L1) +	Gatev & Strahan (2006)
Four-channel demand-deposit core (core)	Disposable income (q/q, L1) +; overnight rate (change, L1) −; saving rate (change, L0) +; household credit (q/q, L1) +	Lucas & Nicolini (2015); Drechsler, Savov & Schnabl (2017); McLeay, Radia & Thomas (2014)
Error-correction model (money demand)	Long run: ln(real demand deposits) on ln real income and the overnight rate (semi-log), deflated by CPI; short run: the core	Lucas & Nicolini (2015); Benati et al. (2016)

Term deposits

Model	Drivers (form, lag)	Source
Rate-gap model	Overnight rate (change, L1) +; overnight rate (change, L2) +; disposable income (q/q, L1) +	Drechsler, Savov & Schnabl (2017)
Portfolio choice	Equity prices (q/q, L1) −; market volatility (change, L0) +; disposable income (q/q, L1) +	Gatev & Strahan (2006); Drechsler, Savov & Schnabl (2017)
Funding need	Mortgage credit (q/q, L0) +; overnight rate (change, L1) +; saving rate (change, L1) +	McLeay, Radia & Thomas (2014)
Four-channel term-deposit core (core)	Disposable income (q/q, L1) +; overnight rate (change, L1) +; equity prices (q/q, L1) −; mortgage credit (q/q, L0) +	Drechsler, Savov & Schnabl (2017); Gatev & Strahan (2006); McLeay, Radia & Thomas (2014)
Error-correction model (term deposits)	Long run: ln(real term deposits) on ln real income and the overnight rate (semi-log, positive), deflated by CPI; short run: the core	Lucas & Nicolini (2015) form, applied to term deposits

The overnight rate enters both cores with a lag of one quarter or more, since posted rates and GIC bookings respond over several quarters. If the credit union's own posted rates are available, the rate gap (term rate minus demand rate) can replace the overnight rate in both models. The regime-switching models use the level of the overnight rate as the threshold variable, since the demand–term mix changes with the rate level.

Investments: drivers, expected signs and models

A credit union's investments are mostly what is left after funding loans: they rise when deposits grow faster than loans and fall when loans outrun deposits, with a floor set by liquidity rules. So the best predictors are the credit union's own deposit and loan forecasts, and macro variables matter mainly through those balances and through the size of the liquidity buffer.

Channel	Driver (series)	Sign	Why	Evidence
Funding inflow	Deposit growth (the demand and term forecasts above)	+	New deposits not lent out are invested	Balance-sheet identity; Kashyap & Stein (2000)
Loan demand	Loan growth (the mortgage, commercial and personal forecasts)	−	Lenders sell or run down securities to fund loans	Bernanke & Blinder (1992); Kashyap & Stein (2000)
Liquidity floor	Net cash outflows under stress (driven by deposit mix)	+	Rules require liquid assets of at least 100% of 30-day stressed outflows	FSRA Liquidity Adequacy Requirements Rule (2022)
Business cycle	Real GDP (FGDPR.ION)	−	Weak loan demand in downturns leaves more funds in securities	Bernanke & Blinder (1992)
Precaution	Market volatility (FVIXC.IC); corporate spread (FRSPREAD.IC)	+	Lenders hoard liquid assets when funding risk rises	Cornett et al. (2011)
Valuation	Change in 5-year Government of Canada yield (map)	−	Higher yields cut the market value of fixed-rate bonds held at fair value	Standard bond pricing
Monetary policy	Overnight rate (FRBOC.IC)	0	Securities fall at once after a tightening and loans only later, so the sign changes with the horizon	Bernanke & Blinder (1992)

The evidence is mostly about why banks hold securities, not about forecasting them:

- **Securities are the first buffer.** After a U.S. tightening, banks first cut securities and only later cut loans (Bernanke & Blinder 1992). Monetary policy moves lending more at banks with fewer securities relative to assets, mainly among small banks (Kashyap & Stein 2000), which is the relevant group for credit unions.
- **Stress builds the buffer.** In 2007–09, banks with more illiquid assets built up liquid assets and cut lending (Cornett et al. 2011).
- **The rule sets the floor.** FSRA's Liquidity Adequacy Requirements Rule took effect on 1 March 2022 with the Credit Unions and Caisses Populaires Act, 2020; it requires a liquidity coverage ratio of at least 100% outside times of stress, plus the NSFR and net cumulative cash flow metrics (FSRA rule page; proposed rule notice). Before 2022 the requirements sat in O. Reg. 237/09, so 2022 is a possible break.

Model	Drivers (form, lag)	Source
Balance-sheet residual	Deposit growth (q/q, L0) +; loan growth (q/q, L0) −	Balance-sheet identity; Kashyap & Stein (2000)
Liquidity buffer	Real GDP (q/q, L1) −; market volatility (change, L0) +; corporate spread (change, L0) +	Bernanke & Blinder (1992); Cornett et al. (2011)
Four-channel investment core (core)	Deposit growth (q/q, L0) +; loan growth (q/q, L0) −; real GDP (q/q, L1) −; 5-year yield (change, L0) −	Kashyap & Stein (2000); Bernanke & Blinder (1992)

Investments should be forecast after deposits and loans, using their forecasts as inputs. Modelled alone from macro data, investments would ignore the identity that ties them to the rest of the balance sheet. The core leaves out an error-correction form: a residual balance has no stable long-run relationship with macro variables, so the long run comes from the deposit and loan error-correction models instead.

Deposits, investments and loans side by side

The overnight rate is the clearest case of one variable with opposite signs across the balance sheet: it lowers demand deposits, raises term deposits and has no prior for investments or business loans. The loan columns repeat the signs from the mortgage, commercial and personal reviews.

Driver	Demand deposits	Term deposits	Investments	Mortgages	Commercial	Personal
Disposable income	+ (core)	+ (core)	0 (via deposits)	+ (core)	0 (indirect)	+ (core)
Real GDP	+	0	− (weak loan demand)	+	+ (core)	+
Overnight rate	− (core)	+ (core)	0 (horizon)	−	0 (horizon)	−
Corporate spread	+ (safety)	0	+ (precaution)	not used	− (core)	0
Unemployment	0	0	0	− (core)	−	− (core)
Equity prices	0	− (core)	0	+	+ (net worth)	0

Drivers used only for deposits and investments:

Driver	Demand deposits	Term deposits	Investments
Inflation	+ (nominal scale)	0	0
Saving rate	+ (core)	+	0
Market volatility	+ (safety)	+	+ (precaution)
Household credit	+ (core)	0	− (via loans)
Mortgage credit	0	+ (core)	− (via loans)
5-year yield, change	0	0	− (core, valuation)

The signs for the inputs to investments (deposit and loan growth) come from the balance-sheet identity, not from macro evidence; the table shows how macro variables reach investments through those balances.

Setup checklist before a real run

The deposit models need two rate series the macro data does not have yet; the investment model needs the deposit and loan forecasts first.

- [ ] Map posted rates for demand and term deposits (the credit union's own, or the Bank of Canada posted rates) so the rate gap can replace the overnight rate.

- [ ] Map a 5-year Government of Canada yield for the investment valuation channel.
- [ ] Decide whether high-interest savings accounts are in demand deposits, and keep the definition fixed over the sample.
- [ ] Split registered and non-registered term deposits if possible: registered GICs are fully insured and behave differently.
- [ ] Add a 2020–21 dummy (pandemic surge) to both deposit models, or start the estimation sample to handle it.
- [ ] Run investments after deposits and loans, feeding in their forecasts; test a 2022 break for the FSRA liquidity rule.
- [ ] Map a population-65+ series for the term-deposit ageing driver, or leave it out.

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Literature Review: Loan Loss Allowances (Mortgage, Commercial, Personal)

Nov 5, 2026 · @Xiaokai

Scope and approach

Allowances rise when the economic outlook worsens, not only when losses arrive: under IFRS 9, the allowance is the expected credit loss on today's book given forward-looking scenarios. So the drivers are the same as for defaults (unemployment, house prices, rates, business failures), but they act earlier and more sharply than under the old incurred-loss rules.

- **Mortgage allowance**: residential mortgages, insured and uninsured. Insured loans carry little expected loss, so the allowance follows the uninsured book.
- **Commercial allowance**: loans to businesses, including commercial real estate, agricultural and small-business loans.
- **Personal loan allowance**: lines of credit, HELOCs, cards, auto and other instalment loans.

Two targets are possible, and the choice changes the signs on loan growth:

- **Allowance balance, quarterly log growth** (the loan-model form). Growth equals loan growth plus the change in the coverage ratio, so loan growth enters with a + sign.
- **Coverage ratio (allowance as a % of loans), quarterly change.** This strips out portfolio size and leaves the credit-risk signal. It is the form most of the provisioning literature uses, and the one recommended here; the allowance balance is then the ratio times the loan forecast.

Every sign below is a sign on the coverage ratio (or on allowance growth, which has the same signs for the macro drivers). As in the loan reviews, sign 0 marks drivers whose sign the literature finds changes with the horizon.

Common framework: expected credit loss and procyclicality

Under IFRS 9 the allowance is probability of default × loss given default × exposure, weighted across scenarios. Performing loans (stage 1) carry 12 months of expected loss; loans whose credit risk has risen significantly (stage 2) carry lifetime expected loss;

impaired loans (stage 3) carry lifetime loss on a defaulted exposure. A worse outlook therefore raises the allowance in two ways: higher PD and LGD for every loan, and loans moving from 12-month to lifetime loss.

The literature agrees on four points that shape the model:

- **Old provisions came too late.** Under incurred-loss rules, banks delayed provisions until downturns had already set in, which magnified the cycle's effect on income and capital (Laeven & Majnoni 2003). Across OECD banks, provisions were higher when GDP growth was low (Bikker & Metzemakers 2005).
- **Expected-loss provisions jump at the start of a downturn.** IFRS 9 and CECL produce larger allowances and sharper on-impact responses to a recession than incurred-loss rules; in a calibrated model, a typical recession raises IFRS 9 provisions on impact by about a third of a bank's capital conservation buffer (Abad & Suarez 2018). Much of the jump comes from loans moving into stage 2.
- **Lending booms raise later losses.** Rapid credit growth is followed, with a long lag, by higher loan losses, because standards loosen in booms (Jiménez & Saurina 2006). Past loan growth therefore carries a + sign on the coverage ratio at long lags, and a − sign at short lags (new loans dilute the ratio).
- **Management judgment matters.** Provisions also reflect smoothing and capital management (Beatty & Liao 2011; Bushman & Williams 2012), and under IFRS 9 they include post-model overlays. A macro model explains the systematic part; overlays and scenario-weight changes show up as residuals.

The 2020 pandemic is the clearest IFRS 9 example: Canada's six largest banks set aside almost $11 billion for credit losses in a single quarter (Q2 2020) (CBC/Reuters), before defaults rose, and released much of it in 2021. Any model estimated on post-2018 data will be dominated by that episode.

Mortgage allowance: drivers and expected signs

Mortgage losses need two things: a borrower who cannot pay, and a home worth less than the loan. Unemployment drives the first and house prices the second. In U.S. household data, losing a job raises the chance of default as much as a 35% fall in home equity (Gerardi et al. 2018). Negative equity is necessary for default but not sufficient: fewer than 10% of Massachusetts owners with negative equity in 1991 lost their homes over the next three years (Foote, Gerardi & Willen 2008). In Canada, mortgage arrears track the unemployment rate closely, follow mortgage rates with a lag, and fall when house prices grow fast (Pugh, Wang & Webley 2026).

Channel	Driver (series)	Sign	Why	Evidence
Ability to pay	Unemployment (FLBR.ION)	+	Job loss is the main trigger of arrears	Gerardi et al. (2018); Pugh, Wang & Webley (2026)
Equity and loss severity	House prices (FHPI.ION)	−	Equity lets borrowers sell instead of default, and lowers loss given default	Foote, Gerardi & Willen (2008); Pugh, Wang & Webley (2026)
Payment shock	5-year mortgage rate (map), lagged	+	Higher rates raise payments at renewal; arrears follow with a lag	Pugh, Wang & Webley (2026)
Nominal income growth	Core or headline inflation (FCPI.ION)	−	Rising nominal incomes shrink real debt burdens; high inflation has coincided with falling arrears	Pugh, Wang & Webley (2026)
Activity	Real GDP (FGDPR.ION)	−	A weaker economy raises expected default in scenarios	Laeven & Majnoni (2003); Bikker & Metzemakers (2005)
Income	Disposable income (FYPD.ION)	−	More income, more capacity to pay	Gerardi et al. (2018)
Household distress	Consumer insolvencies (FBKCPT.ION)	+	Insolvency filings signal rising household defaults	Gross & Souleles (2002b)
Leverage	Debt-to-income (FDTI.IC)	+	More indebted households have less buffer and cut back more in a downturn	Mian, Rao & Sufi (2013)

Channel	Driver (series)	Sign	Why	Evidence
Past lending boom	Mortgage credit (FCRMORT.IC), lagged 8–12 quarters	+	Loans made in booms default more later	Jiménez & Saurina (2006)

Insurance changes the scale, not the signs. CMHC and private mortgage insurance cover lender losses on insured loans, so their expected loss is near zero; the allowance comes almost entirely from uninsured loans. A rising insured share therefore lowers the coverage ratio on the whole book without any change in credit risk.

Commercial allowance: drivers and expected signs

Business loan losses follow firms' sales and their interest burden: they rise when output falls, when credit spreads widen and when rates rise on floating-rate debt, and they are heaviest in commercial real estate. For small Canadian firms on variable rates, higher rates raise debt payments and arrears with a lag, about as much as the general slowdown does (Sekkel, Takamura & Terajima 2024).

Channel	Driver (series)	Sign	Why	Evidence
Firm revenue	Real GDP (FGDPR.ION)	−	Falling sales weaken debt service	Laeven & Majnoni (2003); Bikker & Metzemakers (2005)
Firm failures	Business insolvencies (FBKBUS.IC)	+	Insolvencies are realized defaults	Insolvency is realized default; no separate study needed
Interest burden	Prime rate (FRPRIME.IC), lagged 2–4 quarters	+	Most business loans float at prime; higher rates raise payments and arrears	Sekkel, Takamura & Terajima (2024)

Channel	Driver (series)	Sign	Why	Evidence
Market risk pricing	Corporate spread (FRSPREAD.IC)	+	Wider spreads signal higher expected defaults and feed into scenario PDs	Gilchrist & Zakrajšek (2012)
Collateral	Commercial real estate prices (map)	−	Lower property values raise loss given default on CRE and secured loans	Covas, Rump & Zakrajšek (2014)
Uncertainty	Market volatility (FVIXC.IC)	+	Higher uncertainty raises scenario PDs and stage-2 transfers	Gilchrist & Zakrajšek (2012)
Expectations	Business outlook (FBOS.IC)	−	Weaker outlooks precede weaker cash flow	Leading indicator; not tested directly on losses
Labour market	Unemployment (FLBR.ION)	+	A broad cycle indicator, weaker than GDP for firms	Bikker & Metzemakers (2005)
Past lending boom	Business credit (FCRBUS.IC), lagged 8–12 quarters	+	Loans made in booms default more later	Jiménez & Saurina (2006)

Commercial losses are lumpy and nonlinear. A few large defaults can move a credit union's commercial allowance more than the macro cycle does, and linear models understate losses in a crisis, especially for real estate loans (Covas, Rump & Zakrajšek 2014). A regime or quantile model is worth testing here more than for the household books.

Personal loan allowance: drivers and expected signs

Personal loan losses follow the labour market most closely, because most personal credit is unsecured and defaults turn into losses quickly. Card delinquency and bankruptcy rise with borrowers' risk and debt levels; in U.S. card data, risk measures explain much of the

variation, and the rest points to a falling cost of default over time (Gross & Souleles 2002b).

Channel	Driver (series)	Sign	Why	Evidence
Ability to pay	Unemployment (FLBR.ION)	+	Job loss turns quickly into missed card and loan payments	Gross & Souleles (2002b); Bikker & Metzemakers (2005)
Realized distress	Consumer insolvencies (FBKCPT.ION)	+	Proposals and bankruptcies are write-offs	Gross & Souleles (2002b)
Early delinquency	Card delinquency (FDELCC.IC)	+	Leads defaults by one to two quarters; moves loans to stage 2	Gross & Souleles (2002b)
Interest burden	Prime rate (FRPRIME.IC), lagged	+	Lines of credit and HELOCs float at prime	Sekkel, Takamura & Terajima (2024) (same channel for small firms)
Income	Disposable income (FYPD.ION)	−	More income, more capacity to pay	Gerardi et al. (2018)
Home equity	House prices (FHPI.ION)	−	HELOCs are secured on the home; equity lowers loss given default	Foote, Gerardi & Willen (2008)
Leverage	Debt-to-income (FDTI.IC)	+	Higher debt leaves less room to absorb shocks	Mian, Rao & Sufi (2013)
Past lending boom	Consumer credit (FCRCONS.IC), lagged 8–12 quarters	+	Credit extended in booms goes to riskier borrowers	Jiménez & Saurina (2006)
Activity	Real GDP (FGDPR.ION)	−	Broad cycle indicator	Laeven & Majnoni (2003)

The product mix sets the coverage level. Unsecured cards and lines carry much higher loss given default than HELOCs, so a book dominated by HELOCs has a low coverage ratio that responds to house prices, while a card-heavy book has a high ratio that responds to unemployment. As in the personal loan review, check which products the balance includes before trusting the signs on house prices.

Where the literature disagrees, and Canadian specifics

The main open questions are about timing: how far ahead allowances move, and how much of the move is the model versus management judgment.

- **Rates: payment shock or easing?** Higher rates raise arrears with a lag through payments on floating and renewing loans (Pugh, Wang & Webley 2026; Sekkel, Takamura & Terajima 2024). But rates also rise when the economy is strong and unemployment low, so in a short sample the raw correlation can be negative. The rate therefore enters at lags of 2–4 quarters, alongside unemployment, not alone.
- **Inflation: help or harm?** Moderate inflation erodes real debt and has coincided with falling Canadian mortgage arrears (Pugh, Wang & Webley 2026), but it also brings higher rates. Its sign is negative in the mortgage model only when the rate is in the model too; otherwise it gets no prior.
- **Loan growth: dilution now, losses later.** New loans lower the coverage ratio at once, because they start in stage 1 with 12-month loss; the same boom raises losses two to three years later (Jiménez & Saurina 2006). Short and long lags of loan growth carry opposite signs.
- **Before and after IFRS 9.** Under incurred-loss rules, provisions lagged the cycle (Laeven & Majnoni 2003); under IFRS 9 they lead it (Abad & Suarez 2018). A series that spans 2018 mixes the two regimes, so the response to the same shock differs before and after.

Canadian and credit union specifics:

- **IFRS 9 start date.** IFRS 9 applies to annual periods beginning on or after 1 January 2018; OSFI asked the domestic systemically important banks to adopt it early, from 1 November 2017 (OSFI). Credit unions adopted it from their first fiscal year starting in 2018, so the allowance series needs a break at that date.
- **Short history under the current standard.** Since 2018 the data hold one large shock (2020) and one rate cycle (2022–24). Estimating on a longer incurred-loss history and adding an IFRS 9 step and stage-transfer effect is safer than estimating on post-2018 data alone.

- **Mortgage insurance.** A large insured share (CMHC and private insurers) keeps mortgage coverage ratios low; model the uninsured book's coverage, or include the insured share as a control.
- **Overlays.** Post-model adjustments for events models cannot capture (the pandemic, tariffs) were large in 2020. Record them separately if possible, so the macro model is fitted to the modelled allowance only.

Published models for the workbook

Each segment gets two or three fixed specifications, a four-channel core and a nonlinear variant; the target is the quarterly change in the coverage ratio. Allowances are persistent, so every model also carries the lagged change in the ratio, as in the stress-test satellite models (Covas, Rump & Zakrajšek 2014).

Mortgage allowance

Model	Drivers (form, lag)	Source
Double trigger	Unemployment (change, L1) +; house prices (q/q, L1) −	Foote, Gerardi & Willen (2008); Gerardi et al. (2018)
Canadian arrears model	Unemployment (change, L0) +; house prices (q/q, L1) −; mortgage rate (change, L4) +; inflation (q/q, L1) −	Pugh, Wang & Webley (2026)
Four-channel mortgage-allowance core (core)	Unemployment (change, L1) +; house prices (q/q, L1) −; mortgage rate (change, L3) +; consumer insolvencies (q/q, L0) +	Gerardi et al. (2018); Pugh, Wang & Webley (2026)
Quantile variant	The core, estimated at the 90th percentile	Covas, Rump & Zakrajšek (2014)

Commercial allowance

Model	Drivers (form, lag)	Source
Provisioning cycle	Real GDP (q/q, L0) −; real GDP (q/q, L1) −; unemployment (change, L1) +	Laeven & Majnoni (2003); Bikker & Metzemakers (2005)
Interest burden	Prime rate (change, L2) +; real GDP (q/q, L1) −; business insolvencies (q/q, L0) +	Sekkel, Takamura & Terajima (2024)
Four-channel commercial-allowance core (core)	Real GDP (q/q, L1) −; business insolvencies (q/q, L0) +; prime rate (change, L2) +; corporate spread (change, L1) +	Laeven & Majnoni (2003); Gilchrist & Zakrajšek (2012); Sekkel, Takamura & Terajima (2024)
Regime variant	The core, with real GDP growth as the threshold variable	Covas, Rump & Zakrajšek (2014)

Personal loan allowance

Model	Drivers (form, lag)	Source
Unsecured default	Unemployment (change, L1) +; consumer insolvencies (q/q, L0) +; card delinquency (change, L0) +	Gross & Souleles (2002b)
Rate and leverage	Prime rate (change, L2) +; debt-to-income (change, L1) +; disposable income (q/q, L1) −	Mian, Rao & Sufi (2013); Gerardi et al. (2018)
Four-channel personal-allowance core (core)	Unemployment (change, L1) +; consumer insolvencies (q/q, L0) +; prime rate (change, L2) +; house prices (q/q, L1) −	Gross & Souleles (2002b); Foote, Gerardi & Willen (2008)
Regime variant	The core, with the change in unemployment as the threshold variable	Covas, Rump & Zakrajšek (2014)

An error-correction form is not proposed: the coverage ratio is already stationary in most samples, so there is no long-run level relationship to estimate. If you model the allowance balance instead, use the loan-balance error-correction models for the level and these models for the ratio.

Three allowances side by side

Unemployment drives the two household allowances, and GDP, insolvencies and spreads drive the commercial one; house prices matter for mortgages and HELOC-heavy personal books only. Signs are on the change in the coverage ratio.

Driver	Mortgage	Commercial	Personal
Unemployment	+ (core)	+	+ (core)
Real GDP	−	− (core)	−
House prices	− (core)	0	− (core, via HELOCs)
Commercial real estate prices	0	−	0
Mortgage rate	+ (core, lag 3–4)	0	0
Prime rate	0	+ (core, lag 2)	+ (core, lag 2)
Corporate spread	0	+ (core)	0
Inflation	− (with the rate in the model)	0	0
Disposable income	−	0	−
Consumer insolvencies	+ (core)	0	+ (core)
Business insolvencies	0	+ (core)	0
Card delinquency	0	0	+
Debt-to-income	+	0	+
Market volatility	0	+	0

Driver	Mortgage	Commercial	Personal
Business outlook	0	−	0
Own loan growth, lag 8–12	+	+	+

The signs mirror the loan-balance reviews for the household variables: what raises borrowing capacity (income, house prices) lowers expected loss, and what weakens it (unemployment, insolvencies, higher rates) raises it.

Setup checklist before a real run

The allowance models need a target defined consistently across the IFRS 9 break and a few series the macro data lacks.

- [ ] Build the coverage ratio (allowance ÷ gross loans) for each segment, and decide whether to use it or allowance growth as the target.
- [ ] Split the mortgage book into insured and uninsured, or add the insured share as a control.
- [ ] Add an IFRS 9 step dummy at the credit union's first 2018 fiscal year, and a 2020 dummy if overlays cannot be separated.
- [ ] Map a 5-year mortgage rate and a commercial real estate price index.
- [ ] Check which products the personal balance includes (HELOCs, cards, auto); the house-price sign depends on the HELOC share.
- [ ] Feed the scenario paths into the same models the allowance forecast uses, so the allowance and loss forecasts stay consistent with the loss model.

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Literature Review: Loan and Deposit Interest Rates

Nov 5, 2026 · @Xiaokai

Scope and approach

Retail rates are a market benchmark plus a margin: each rate follows the market rate of matching maturity, fully in the long run for most loans and only partly for demand deposits, and the margin moves with funding costs, credit risk and competition. So the core of every model is the pass-through from one benchmark, and the other drivers explain the margin.

Rate	Benchmark it follows	Long-run pass-through
Fixed mortgage rate (5-year)	5-year Government of Canada yield	Close to full
Variable mortgage rate, HELOC rate	Prime rate, so the overnight rate	Full, by contract
Commercial loan rate	Prime rate (floating) or matching-term yield (fixed)	Close to full
Personal loan and line of credit rate	Prime rate; cards are sticky	Full for lines; low for cards
Demand-deposit rate (chequing, savings)	Overnight rate	Partial (well below one)
Term-deposit (GIC) rate	Government yield of matching term	Close to full

The target is the quarterly change in each rate (percentage points), not log growth, since rates can sit near zero. Signs below are on that change. Where the macro data has no series for the benchmark (5-year yield, GIC rates), the table says what to map.

Common framework: pass-through and the cost of funds

The standard model is an error-correction equation: in the long run the retail rate equals a margin plus β times the benchmark rate, and in the short run it closes part of any gap each quarter. β is the long-run pass-through; the speed of adjustment says how fast it arrives.

$$\Delta r_t = \alpha\,(r_{t-1} - \mu - \beta\, m_{t-1}) + \gamma\, \Delta m_t + \delta' \Delta x_t + \varepsilon_t$$

Here r is the retail rate, m the benchmark and x the drivers of the margin. Four findings shape the specification:

- **Pass-through is slow and incomplete in the short run, close to complete in the long run for loans.** Lending rates adjust to market rates over several quarters, with long-run pass-through near one for mortgages and business loans and lower for deposits (Cottarelli & Kourelis 1994; de Bondt 2005).
- **Bank features change the speed, not the end point.** Differences across banks in pass-through, tied to liquidity, capital and relationship lending, appear only in the short run; long-run elasticities are the same (Gambacorta 2008).
- **Funding costs, not the policy rate, anchor lending rates.** After 2008 lending rates stopped tracking policy rates, but stayed tied to banks' weighted average cost of liabilities (Illes, Lombardi & Mizen 2019). For a credit union funded by deposits, the cost of its deposit book is the natural benchmark for lending margins.
- **Market power makes deposit rates sticky and asymmetric.** Deposit rates rise slowly when market rates rise and fall quickly when they fall, more so in concentrated markets (Neumark & Sharpe 1992). The resulting low deposit beta is what lets banks fund long-term fixed-rate loans without large interest-rate risk (Drechsler, Savov & Schnabl 2021).

Mortgage rates: drivers and expected signs

Canadian fixed mortgage rates follow the 5-year Government of Canada yield, and variable rates follow prime, which moves with the overnight rate. Using posted rates, pass-through from market rates looks incomplete; using the discounted rates borrowers actually pay, it is complete, with some asymmetry in timing (Allen & McVanel 2009). Discounting therefore affects how fast and how much policy changes reach mortgage rates (Allen 2011).

Channel	Driver (series)	Sign	Why	Evidence
Benchmark, fixed	5-year Government of Canada yield (map), change	+	Fixed rates are priced as the matching-term yield plus a margin	Allen & McVanel (2009); de Bondt (2005)
Benchmark, variable	Prime rate (FRPRIME.IC), change	+	Variable mortgages and HELOCs are prime plus or minus a set spread	Allen & McVanel (2009)
Funding cost	Lender's cost of deposits or mortgage-bond spread (map)	+	Margins track the cost of the lender's own funding	Illes, Lombardi & Mizen (2019)
Market risk	Corporate spread (FRSPREAD.IC)	+	Wider spreads raise wholesale and term funding costs	Illes, Lombardi & Mizen (2019)
Credit risk	Unemployment (FLBR.ION)	0	Uninsured margins may widen in a downturn, but insured loans carry little credit risk and weak demand pushes margins down	No settled evidence
Competition	Market concentration, posted-to-discount gap (map if available)	+	Less competition, smaller discounts	Allen, Clark & Houde (2014)

Insured mortgages are funded partly through Canada Mortgage Bonds, whose 5-year yields sat only about 25 basis points above Government of Canada yields in 2024 (Desjardins 2024). Credit unions fund more from member deposits, so their margin over the 5-year yield moves more with their own deposit costs.

Commercial and personal loan rates: drivers and expected signs

Commercial and personal loan rates are mostly prime plus a margin, so the overnight rate passes through almost fully; the margin then widens with borrower risk. Credit cards are the exception: their rates barely move with market rates, because borrowers who expect to repay ignore the rate and those who carry balances rarely switch (Ausubel 1991; Calem & Mester 1995).

Commercial loan rate

Channel	Driver (series)	Sign	Why	Evidence
Benchmark	Prime rate (FRPRIME.IC), change	+	Most business loans float at prime plus a margin	Cottarelli & Kourelis (1994); de Bondt (2005)
Fixed-rate term loans	Matching-term Government of Canada yield (map), change	+	Fixed business loans price off the term yield	de Bondt (2005)
Credit risk	Corporate spread (FRSPREAD.IC)	+	Market pricing of default risk feeds into loan margins	Gilchrist & Zakrajšek (2012)
Realized risk	Business insolvencies (FBKBUS.IC)	+	Higher defaults lead lenders to raise margins and tighten terms	Gilchrist & Zakrajšek (2012)
Funding cost	Lender's average cost of liabilities (map)	+	Margins follow the cost of funds	Illes, Lombardi & Mizen (2019)
Lender balance sheet	Lender liquidity and capital	− (on the speed of pass-through)	Liquid, well-capitalized lenders smooth rates and pass changes through more slowly	Gambacorta (2008)

Personal loan rate

Channel	Driver (series)	Sign	Why	Evidence
Benchmark	Prime rate (FRPRIME.IC), change	+	Lines of credit and HELOCs float at prime	Cottarelli & Kourelis (1994)

Channel	Driver (series)	Sign	Why	Evidence
Credit card rate	Overnight rate (FRBOC.IC), change	0	Card rates are sticky and barely follow market rates	Ausubel (1991); Calem & Mester (1995)
Borrower risk	Consumer insolvencies (FBKCPT.ION); unemployment (FLBR.ION)	+	Higher default risk widens margins on unsecured credit	Gross & Souleles (2002b)
Funding cost	Lender's average cost of liabilities (map)	+	Margins follow the cost of funds	Illes, Lombardi & Mizen (2019)

For a credit union's average personal loan rate, the product mix matters as much as the drivers: a book that shifts toward cards raises the average rate and lowers its sensitivity to prime, and a shift toward HELOCs does the opposite.

Deposit rates (demand and term): drivers and expected signs

Demand-deposit rates pass through a fraction of policy changes, slowly and more slowly upward; GIC rates follow government yields of matching term almost fully. In the U.S., a 100 bp rise in the policy rate widens deposit spreads by about 61 bp on average, so deposit rates rise by well under half the move (Drechsler, Savov & Schnabl 2017); by the end of 2022 the cumulative beta on interest-bearing deposits had risen to almost 0.4 (Liberty Street Economics). Betas also rise as a hiking cycle goes on and rates get higher.

Demand-deposit rate

Channel	Driver (series)	Sign	Why	Evidence
Benchmark	Overnight rate (FRBOC.IC), change	+ (beta well below 1)	Banks raise deposit rates by only part of a policy move	Drechsler, Savov & Schnabl (2017); Hannan & Berger (1991)
Asymmetry	Overnight rate rises versus cuts	+ up slower than down	Rates adjust upward more slowly than downward	Neumark & Sharpe (1992)

Channel	Driver (series)	Sign	Why	Evidence
Rate level	Level of the overnight rate	+ (on the beta)	Betas rise when rates are high, as depositors move money	Liberty Street Economics (2023)
Funding need	Loan growth minus deposit growth (own data)	+	Lenders short of deposits bid for them	Gambacorta (2008) (deposit pass-through depends on liability structure)
Competition	Online-bank and competitor savings rates (map)	+	Rival rates set the market price of savings	Neumark & Sharpe (1992)

Term-deposit (GIC) rate

Channel	Driver (series)	Sign	Why	Evidence
Benchmark	Government of Canada yield of matching term (map), change	+ (close to 1)	GICs compete directly with bonds of the same term	de Bondt (2005)
Policy rate	Overnight rate (FRBOC.IC), change	+	Short GICs track the expected path of the overnight rate	Drechsler, Savov & Schnabl (2017)
Funding need	Loan growth minus deposit growth (own data)	+	GICs are the main tool to raise term funding quickly	Gambacorta (2008)
Market stress	Corporate spread (FRSPREAD.IC)	0	Wholesale funding gets dearer (+), but insured deposits flow in on their own (−)	Gatev & Strahan (2006)

The gap between the GIC rate and the demand-deposit rate is what moves money between the two balances (the deposits review). Modelling both rates with the same benchmark set keeps the rate gap consistent with the balance forecasts.

Where the literature disagrees, and Canadian specifics

The open questions are which benchmark to use and whether pass-through is stable over time.

- **Policy rate or cost of funds?** Policy-rate pass-through broke down after 2008 in Europe, while the link to banks' actual funding costs held (Illes, Lombardi & Mizen 2019). For a credit union, the overnight rate is the natural scenario input, but the margin model should include its deposit cost so that a change in the deposit mix shows up in lending rates.
- **Is the deposit beta a constant?** Betas are low early in a hiking cycle and rise as rates climb (Liberty Street Economics 2023), and they differ between hikes and cuts (Neumark & Sharpe 1992). A single linear pass-through will understate deposit costs late in a long tightening; a regime model with the rate level as threshold, or separate up and down coefficients, handles this.
- **Posted or effective rates?** Pass-through looks incomplete with posted mortgage rates and complete with discounted transaction rates (Allen & McVanel 2009). Model the rate the credit union actually earns or pays (average effective rate on the book, or rate on new business), not posted rates.
- **Book rate or new-business rate?** The average rate on the stock moves slowly, because fixed-rate loans and GICs reprice only at maturity. A five-year term means about a fifth of the book reprices each year, so the stock rate needs a repricing lag structure, not just a pass-through coefficient.

Canadian and credit union specifics:

- **Prime and variable rates.** Variable mortgages, HELOCs and most business lines move with prime, which in turn moves with the overnight rate within days of each Bank of Canada decision.
- **Fixed mortgages follow bonds, not the Bank.** Five-year fixed rates follow the 5-year Government of Canada yield, so they fell much less than variable rates after the cuts that began in June 2024 (Desjardins 2025).
- **Deposit-funded lenders.** Credit unions fund mostly with member deposits rather than wholesale or mortgage-bond funding, so their lending margins depend more on their deposit beta and GIC pricing than the large banks' do.

Published models for the workbook

Every rate gets the same three forms: a pass-through error-correction model on its benchmark, a margin model, and a four-channel core; the error-correction model is the primary specification here, since pass-through is the established structure.

Rate	Error-correction model (long run)	Four-driver core (form, lag, sign)	Source
Fixed mortgage	Rate on the 5-year GoC yield (level)	5-year yield (change, L0) +; 5-year yield (change, L1) +; overnight rate (change, L1) +; corporate spread (change, L1) +	Allen & McVanel (2009); de Bondt (2005); Illes, Lombardi & Mizen (2019)
Variable mortgage, HELOC	Rate on prime (level)	Prime (change, L0) +; overnight rate (change, L1) +; 3-month T-bill (change, L0) +; corporate spread (change, L1) +	Allen & McVanel (2009)
Commercial loan	Rate on prime (level)	Prime (change, L0) +; corporate spread (change, L1) +; business insolvencies (q/q, L1) +; 2-year yield (change, L0) +	Cottarelli & Kourelis (1994); Gilchrist & Zakrajšek (2012); de Bondt (2005)
Personal loan	Rate on prime (level)	Prime (change, L0) +; consumer insolvencies (q/q, L1) +; unemployment (change, L1) +; card delinquency (change, L0) +	Cottarelli & Kourelis (1994); Gross & Souleles (2002b)

Rate	Error-correction model (long run)	Four-driver core (form, lag, sign)	Source
Demand deposit	Rate on the overnight rate (level; $\beta < 1$ estimated)	Overnight rate (change, L0) +; overnight rate (change, L1) +; 3-month T-bill (change, L0) +; mortgage credit (q/q, L0) + (funding-need proxy)	Drechsler, Savov & Schnabl (2017); Hannan & Berger (1991); Gambacorta (2008)
Term deposit (GIC)	Rate on the 2-year GoC yield (level)	2-year yield (change, L0) +; 5-year yield (change, L0) +; overnight rate (change, L0) +; mortgage credit (q/q, L0) +	de Bondt (2005); Gambacorta (2008)

The cores use only drivers that have a series in the macro data. The lender's deposit cost and competitor rates sit in separate margin models ("funding-cost margin", "competition and funding need") until you map them; the pipeline skips an unmapped driver and notes it in the log. Separate coefficients for rate rises and cuts, and a beta that rises with the rate level, are handled by the regime models rather than the core.

The regime-switching models use the level of the overnight rate as the threshold variable for every rate, since pass-through, deposit betas and discounting all change with the rate level. For the average rate on the book rather than on new business, feed the new-business rate from these models into a repricing schedule built from the book's maturity profile.

All rates side by side

The benchmark sets the level of each rate and differs by product; credit-risk variables matter only for lending rates, and funding-need variables matter most for deposit rates. Signs are on the quarterly change in each rate.

Driver	Fixed mortgage	Variable mortgage, HELOC	Commercial	Personal	Demand deposit	GIC
Overnight rate	+ (L1)	+ (via prime)	+ (via prime)	+ (via prime; 0 for cards)	+ (core, β < 1)	+
Prime rate	0	+ (β = 1)	+ (core)	+ (core)	0	0
5-year GoC yield	+ (core, β ≈ 1)	0	+ (fixed loans)	0	0	+ (5-year GICs)
Corporate spread	+	+	+ (core)	0	0	0
Business insolvencies	0	0	+ (core)	0	0	0
Consumer insolvencies	0	0	0	+ (core)	0	0
Unemployment	0	0	0	+	0	0
Lender's deposit cost	+ (margin model)	0	+ (margin model)	+	target	target
Funding gap (loans minus deposits)	0	0	0	0	+ (core)	+ (core)
Competitor rates	+	+	0	0	+ (core)	+ (core)

The deposit-cost row links the models: deposit rates feed the lender's cost of funds, which feeds lending margins. Forecast deposit rates first, then lending rates, so net interest margin scenarios stay consistent.

Setup checklist before a real run

The rate models need benchmark yields the macro data does not hold and a clear choice between book and new-business rates.

- [ ] Map Government of Canada yields at 1, 2, 3 and 5 years (Bank of Canada series) for the fixed mortgage, fixed commercial and GIC models.
- [ ] Decide for each product whether the target is the new-business rate or the average rate on the book; if the book, build the repricing schedule from the maturity profile.
- [ ] Use effective (discounted) rates, not posted rates, for mortgages and GICs.
- [ ] Build the credit union's average deposit cost from the demand and term rates and balances, so it can enter the lending-rate cores.

- [ ] Find a competitor rate series (big-bank posted or online-bank savings and GIC rates) for the deposit models, or drop that driver.
- [ ] Allow separate coefficients for rate rises and cuts in the deposit-rate models, and test a regime split on the overnight-rate level.

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Literature Review: Income, Losses and Capital

Nov 12, 2026 · @Xiaokai

Scope and approach

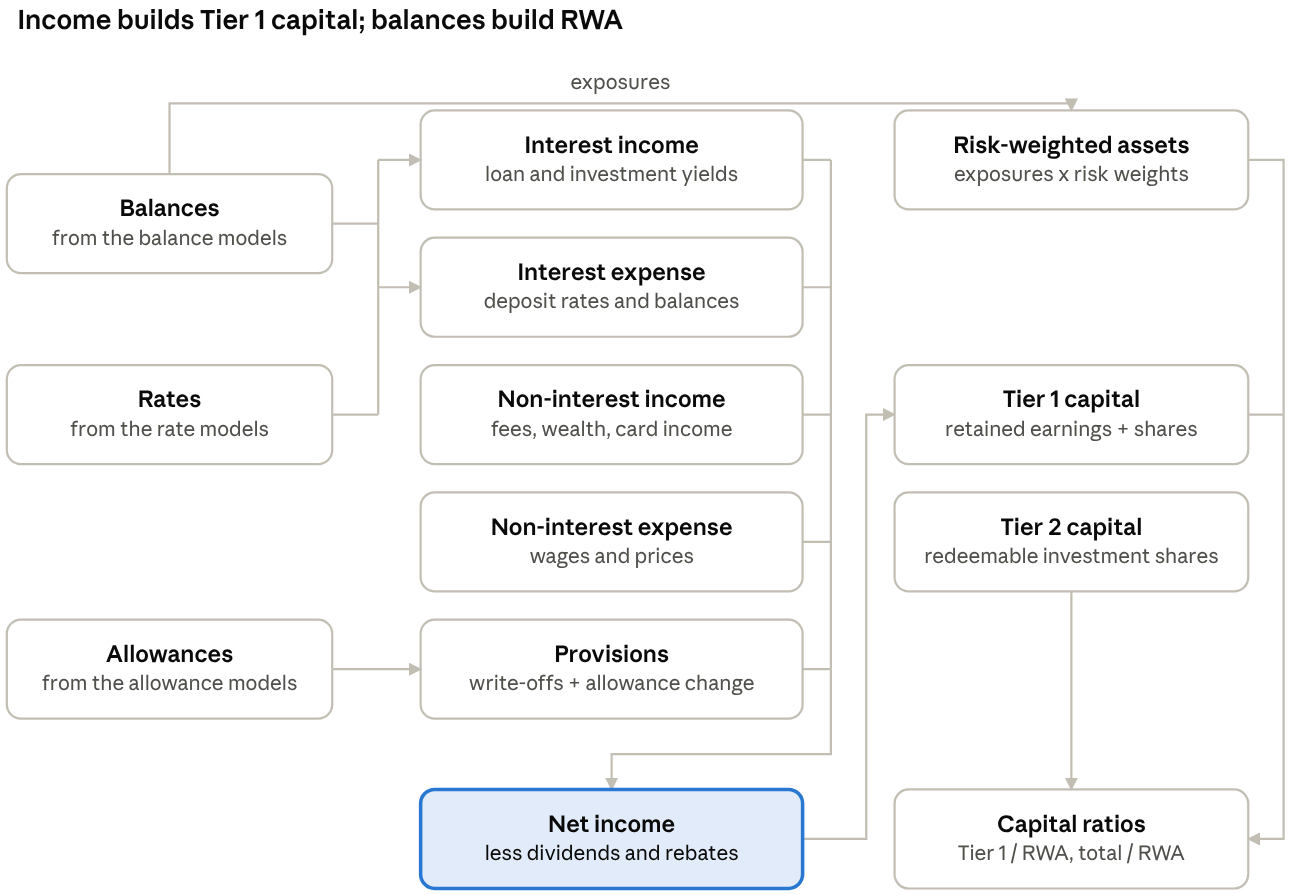
Income, losses and capital are mostly built from the balances and rates already forecast, so this review sets out what each item depends on beyond them. Each item gets the same treatment as the balance reviews: channels, a driver series, an expected sign, and the evidence.

Item	What it covers	Target form
Interest income	Interest on loans and investments	Quarterly log growth
Interest expense	Interest on deposits and borrowings	Quarterly log growth
Net interest margin	Net interest income ÷ average assets, annualized (%)	Quarterly change, pp
Non-interest income	Fees, service charges, wealth and insurance commissions, gains	Quarterly log growth
Non-interest expense	Salaries, premises, IT, other operating costs	Quarterly log growth
Provisions for credit losses	Provision ÷ average loans, annualized (%)	Quarterly change, pp
Net write-offs	Write-offs less recoveries ÷ average loans, annualized (%)	Quarterly change, pp
Risk-weighted assets	Credit plus operational risk-weighted assets	Quarterly log growth
Tier 1 capital	Retained earnings, qualifying investment and membership shares, less deductions	Quarterly log growth
Tier 2 capital	Investment shares redeemable within 12 months and other qualifying instruments	Quarterly log growth

The capital ratios are not separate targets. FSRA's rule sets minimums of 6.5% for Tier 1 capital, 8.0% for total capital and 10.5% for total supervisory capital (with the 2.5% conservation buffer), all as shares of RWA, plus a 3.0% leverage ratio; Tier 1 must include retained earnings of at least 3.0% of RWA (FSRA, Rule 2021-002 notice). Each ratio is a forecast numerator over a forecast denominator, so it follows from the targets above.

As before, sign 0 means no prior, not no effect, and keeps a series out of the data-driven screen.

How the items connect



how the forecast items connect · inputs, income statement, capital

Balances and rates set interest income and expense; allowances set provisions; net income after dividends becomes retained earnings in Tier 1, while lending sets RWA. The capital ratios are then Tier 1 and total capital over RWA.

Net interest income: interest income, interest expense and the margin

Interest income is loan and investment balances times their yields; interest expense is deposit and borrowing balances times their rates. Higher and steeper rate structures raise margins: across 109 large banks in 1995–2012, the level of short rates and the slope of the yield curve were positively related to profitability, with the net interest income effect dominating and the effect largest when rates are low (Borio, Gambacorta & Hofmann 2017). In 3,385 banks from 47 countries, a one-point drop in rates lowered the margin by 8

bp, and by 20 bp at low rates (Claessens, Coleman & Donnelly 2018). Because deposit rates are sticky, banks' margins have been very stable over the rate cycle on average (Drechsler, Savov & Schnabl 2021).

Item	Driver (series)	Sign	Why	Evidence
Interest income	Loan growth (household and business credit)	+	More earning assets	Balance identity; Fed supervisory PPNR models
Interest income	Prime rate (FRPRIME.IC)	+	Floating-rate loans and HELOCs reprice at once	Borio, Gambacorta & Hofmann (2017)
Interest income	5-year GoC yield (FRGOV5Y.IC)	+	Fixed-rate loans and investments reprice at renewal	Borio, Gambacorta & Hofmann (2017)
Interest expense	Overnight rate (FRBOC.IC)	+	Deposit rates follow with a beta below one	Drechsler, Savov & Schnabl (2017)
Interest expense	2-year GoC yield (FRGOV2Y.IC)	+	GIC rates track term yields	de Bondt (2005)
Interest expense	Deposit growth (M2 proxy)	+	More interest-bearing liabilities	Balance identity
Net interest margin	Overnight rate level (FRBOC.IC)	+	Deposit franchise: the spread on sticky deposits widens with rates	Claessens, Coleman & Donnelly (2018); Drechsler, Savov & Schnabl (2017)
Net interest margin	Term spread (10-year minus overnight)	+	Lend long, fund short	Borio, Gambacorta & Hofmann (2017)
Net interest margin	Change in the overnight rate	− short run	Deposit costs on GICs rise before fixed-rate assets reprice for a credit union with a short-funded fixed book	Repricing gap; English (2002)

For a credit union the timing matters: most mortgages are five-year fixed, while GICs and savings reprice faster, so a rate rise first squeezes the margin and widens it only as mortgages renew. FSRA reports Ontario sector net interest income of 1.48% of average assets in 1Q-2024 and 1.69% in 2Q-2025 (FSRA 1Q-2024; FSRA 2Q-2025), a rise that fits mortgages repricing after the 2022–23 hikes, though the reports do not break it down.

Non-interest income and non-interest expense

Non-interest income follows transaction volumes, wealth and housing activity, and it is more volatile than net interest income, not a hedge against it. In U.S. banks it is quite volatile and increasingly correlated with net interest income, so the shift toward it brought few diversification benefits (Stiroh 2004). Non-interest expense is mostly salaries and premises, so it follows wages and inflation with little cyclical swing. The Federal Reserve's supervisory stress test models both as parts of pre-provision net revenue, defined as net interest income plus non-interest income minus non-interest expense, and relates each component to macroeconomic variables (Federal Reserve 2025).

Item	Driver (series)	Sign	Why	Evidence
Non-interest income	Equity prices (FSTSX.IC)	+	Wealth-management and mutual-fund fees scale with assets under administration	Stiroh (2004)
Non-interest income	Consumption, retail sales (FCR.IC, FRS.ION)	+	Card interchange and account service charges follow transactions	Stiroh (2004)
Non-interest income	Existing home sales (FHSEXIST.ION)	+	Mortgage, insurance and penalty fees follow housing turnover	Fed supervisory PPNR models
Non-interest income	Market volatility (FVIXC.IC)	−	Lower fund flows and securities gains in stress	Stiroh (2004)
Non-interest income	Real GDP (FGDPR.ION)	+	Broad activity	Borio, Gambacorta & Hofmann (2017)

Item	Driver (series)	Sign	Why	Evidence
Non-interest expense	Average hourly wages (FWAGE.ION)	+	Salaries are the largest cost	Fed supervisory PPNR models
Non-interest expense	Inflation (FCPI.ION)	+	Premises, IT and services costs rise with prices	Fed supervisory PPNR models
Non-interest expense	Asset or loan growth	+	Larger books need more staff and systems	Balance identity

The efficiency ratio (non-interest expense over revenue) is high in the Ontario sector: 74.2% in 1Q-2026, against 82.5% in 4Q-2025, before dividends and rebates (FSRA 1Q-2026). Small changes in revenue therefore move net income a lot, which is why expenses need their own model rather than a fixed share.

Losses: provisions for credit losses and net write-offs

Provisions equal net write-offs plus the change in the allowance, so once the allowances and write-offs are forecast, provisions follow; the Federal Reserve uses the same identity in its stress test (Federal Reserve 2025). Write-offs lag the cycle, since a loan is written off only after it defaults and collateral is realized; provisions under IFRS 9 lead it, because the allowance part moves with the outlook (the allowances review).

$$\text{PCL}_t = \text{NCO}_t + (\text{Allowance}_t - \text{Allowance}_{t-1})$$

Item	Driver (series)	Sign	Why	Evidence
Net write-offs	Unemployment (FLBR.ION), lagged	+	Defaults follow job loss	Gerardi et al. (2018); Gross & Souleles (2002b)
Net write-offs	House prices (FHPI.ION), lagged	−	Collateral covers more of a defaulted mortgage or HELOC	Foote, Gerardi & Willen (2008)

Item	Driver (series)	Sign	Why	Evidence
Net write-offs	Business insolvencies (FBKBUS.IC)	+	Commercial defaults	Covas, Rump & Zakrajšek (2014)
Net write-offs	Consumer insolvencies (FBKCPT.ION)	+	Personal-loan write-offs	Gross & Souleles (2002b)
Provisions	Real GDP (FGDPR.ION)	−	Provisions rise when growth is low	Laeven & Majnoni (2003); Bikker & Metzemakers (2005)
Provisions	Unemployment (FLBR.ION)	+	Higher scenario PDs and stage-2 transfers	Abad & Suarez (2018)
Provisions	Prime rate (FRPRIME.IC), lagged	+	Payment burden on floating-rate loans	Sekkel, Takamura & Terajima (2024)
Provisions	Corporate spread (FRSPREAD.IC)	+	Market pricing of default risk	Gilchrist & Zakrajšek (2012)

Losses are skewed: linear models fitted to quiet periods understate them in a crisis, especially for real estate, while quantile models capture the tail (Covas, Rump & Zakrajšek 2014). In the Ontario sector, loan costs were only 0.04% of average assets in 1Q-2024 (FSRA 1Q-2024), so the history holds few large-loss quarters to estimate from.

Risk-weighted assets

For a credit union on fixed regulatory risk weights, RWA follows the size and mix of the balance sheet, not the credit cycle. FSRA's rule computes RWA as credit risk-weighted assets plus operational risk under the basic indicator approach, with assets assigned to nine risk-weight categories from 0% to 1250% (FSRA technical briefing 2022). The procyclicality the literature worries about comes from model-based weights, which rise in downturns: Basel II's internal-ratings approach is more procyclical than Basel I (Repullo & Suarez 2013), and model-based charges rose and lending fell after a credit-risk shock in Germany (Behn, Haselmann & Wachtel 2016). With fixed weights, the cycle reaches RWA mainly through loan growth and the share of loans that become past due.

$$\text{RWA}_t = \sum_k w_k \, E_{k,t} + \text{Operational RWA}_t$$

Channel	Driver (series)	Sign	Why	Evidence
Commercial lending	Business credit (FCRBUS.IC)	+	Business loans carry the highest standard weights	Usual standardized weights; confirm in the rule's Table 2
Household lending	Household and mortgage credit (FCRHH.IC, FCRMORT.IC)	+	Uninsured mortgages and personal loans add RWA; insured mortgages add little	Usual standardized weights; confirm in the rule's Table 2
Liquidity holdings	Deposit growth (M2 proxy)	0	Deposits placed in low-weight government securities add little RWA	Usual standardized weights; confirm in the rule's Table 2
Operational risk	Gross income (net interest + non-interest income)	+	The basic indicator approach scales operational RWA with gross income	FSRA technical briefing (2022)
Credit deterioration	Unemployment, business insolvencies	+ small	Past-due exposures move to higher weights	Behn, Haselmann & Wachtel (2016) (mechanism)
Housing collateral	House prices (FHPI.ION)	0	Matters only where weights depend on loan-to-value	FSRA (2023), risk-weight review

FSRA is reviewing mortgage risk weights to use loan-to-value, debt service and amortization, possibly with finer metrics (FSRA TAC, November 2023). If adopted, house prices and debt-service ratios would enter RWA directly, so the sign on house prices would turn negative.

Tier 1 and Tier 2 capital

A credit union cannot raise common equity in public markets, so Tier 1 capital grows mainly through retained earnings, and the macro drivers of Tier 1 are the drivers of net income. Under FSRA's rule, Tier 1 predominantly includes retained earnings and investment shares not redeemable within 12 months; Tier 2 predominantly includes investment shares redeemable within 12 months and other qualifying securities (FSRA, Rule 2021-002 notice). Banks worldwide also rebuilt capital after 2008 mainly through retained earnings, helped by lower payouts and wider lending spreads (Cohen & Scatigna 2016).

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\text{Tier 1}_t = \text{Tier 1}_{t-1} + \text{Net income}_t - \text{Dividends and rebates}_t + \Delta\text{Qualifying shares}_t + \Delta\text{AOCI}_t - \Delta\text{Deductions}_t
```

Item	Driver (series)	Sign	Why	Evidence
Tier 1	Overnight rate level (FRBOC.IC)	+	Higher margins raise net income and retained earnings	Borio, Gambacorta & Hofmann (2017); Cohen & Scatigna (2016)
Tier 1	Unemployment (FLBR.ION)	−	Higher provisions cut net income	Hirtle et al. (2016)
Tier 1	Real GDP (FGDPR.ION)	+	Income and fees grow, provisions fall	Hirtle et al. (2016)
Tier 1	5-year GoC yield, change (FRGOV5Y.IC)	−	Unrealized losses on fair-value securities pass through AOCI, which FSRA counts in Tier 1	FSRA Q&A (2022)
Tier 1	House prices (FHPI.ION)	+	Lower mortgage and HELOC losses	Foote, Gerardi & Willen (2008)
Tier 2	Investment shares entering the 12-month redemption window	+	Shares move from Tier 1 to Tier 2 as they become redeemable	FSRA (2022), stakeholder summary
Tier 2	Disposable income (FYPD.ION)	+	Members' capacity to buy new shares	No direct study

Item	Driver (series)	Sign	Why	Evidence
Tier 2	Equity prices (FSTSX.IC)	–	Competing investments for members' savings	No direct study

Investment shares are a large part of sector capital: $2.8 billion, or 41.9% of capital, in 1Q-2023, when retained earnings grew 2.8% against 8.3% asset growth (FSRA 1Q-2023). Credit unions may redeem only 10% of a class's shares outstanding in a year, and shares eligible for redemption in the next 12 months are deducted from Tier 1 and assigned to Tier 2 (FSRA stakeholder summary). Tier 2 is therefore driven more by share schedules than by the economy, and is best projected from the share register with the macro model as a check.

Where the literature disagrees, and Canadian specifics

The main open question is how much of the rate effect on income survives once the balance sheet adjusts.

- **Do higher rates raise or lower profits?** Cross-country evidence finds higher, steeper rate structures raise margins and profitability, more so at low rates (Borio, Gambacorta & Hofmann 2017; Claessens, Coleman & Donnelly 2018). But U.S. banks' net interest margins have been stable across rate cycles because deposit costs and asset yields are matched (Drechsler, Savov & Schnabl 2021). For a credit union with five-year fixed mortgages and faster-repricing GICs, the short-run effect of a rate rise can be negative before it turns positive, so the margin model needs both the level and the change of the policy rate.
- **Does non-interest income stabilize earnings?** It is more volatile and moves with net interest income, so it adds little diversification (Stiroh 2004). Treat it as cyclical, not as a cushion.
- **How procyclical is capital?** With risk-sensitive weights, required capital rises in downturns while losses eat into capital (Repullo & Suarez 2013; Behn, Haselmann & Wachtel 2016). With standardized weights, the procyclicality runs mainly through earnings: provisions cut retained earnings while RWA barely moves.
- **Losses: linear or not?** Linear loss models fit quiet periods and understate crisis losses; quantile or regime models do better in the tail (Covas, Rump & Zakrajšek 2014).

Canadian and credit union specifics:

- **Capital rule since March 2022.** FSRA's Capital Adequacy Requirements Rule raised the minimum total supervisory capital ratio to 10.5% (from 8% under the old rule) and added a 3.0% retained-earnings-to-RWA floor (FSRA, Rule 2021-002 notice; DUCA submission). Series spanning 2022 carry a definitional break in RWA and in the Tier 1/Tier 2 split.
- **Sector levels.** The Ontario sector's risk-weighted capital ratio was 13.78% in 4Q-2025 and its leverage ratio 6.86% in 3Q-2025 (FSRA 4Q-2025; FSRA 3Q-2025), well above the minimums.
- **Retained earnings lag assets.** FSRA has repeatedly reported retained earnings growing more slowly than assets, for example 7.6% against 9.8% year over year in one third-quarter report (FSRA Q3 release), which keeps capital ratios tied to profitability.
- **AOCI counts.** FSRA confirmed that gains and losses in accumulated other comprehensive income stay in Tier 1 (FSRA Q&A 2022), so bond-yield changes reach Tier 1 directly through fair-value securities.

Published models for the workbooks

Each item has one core model, used by the hybrids, regime and network models, and one or two published specifications alongside it. The table shows the cores; the other models are in each workbook's models sheet.

Item	Core drivers (form, lag, sign)	Error-correction model	Regime threshold	Source
Interest income	Household credit (q/q, L0) +; business credit (q/q, L0) +; prime (change, L0) +; 5-year yield (change, L2) +	Yes: real household credit and prime level	Overnight level	Federal Reserve (2025); Borio, Gambacorta & Hofmann (2017)
Interest expense	Deposits, M2 proxy (q/q, L0) +; overnight (change, L0) +; overnight (change, L1) +; 2-year yield (change, L0) +	Yes: real deposits and overnight level	Overnight level	Drechsler, Savov & Schnabl (2017); de Bondt (2005)

Item	Core drivers (form, lag, sign)	Error-correction model	Regime threshold	Source
Net interest margin	Overnight level (L0) +; overnight change (L0) −; 5-year yield (change, L4) +; 10-year yield (change, L0) +	No (ratio)	Overnight level	Borio, Gambacorta & Hofmann (2017); Claessens, Coleman & Donnelly (2018); English (2002)
Non-interest income	TSX (q/q, L0) +; consumption (q/q, L0) +; home sales (q/q, L0) +; VIXC (q/q, L0) −	Yes: real consumption and real equity prices	GDP growth	Stiroh (2004); Federal Reserve (2025)
Non-interest expense	Wages (q/q, L0) +; CPI (q/q, L0) +; household credit (q/q, L1) +	Yes: real wages and real credit	Inflation	Federal Reserve (2025)
Provision rate	Unemployment (change, L1) +; GDP (q/q, L0) −; corporate spread (change, L1) +; prime (change, L2) +	No (ratio)	Unemployment change	Abad & Suarez (2018); Laeven & Majnoni (2003); Sekkel, Takamura & Terajima (2024)
Net write-off rate	Unemployment (change, L2) +; house prices (q/q, L2) −; business insolvencies (q/q, L1) +; consumer insolvencies (q/q, L1) +	No (ratio)	Unemployment change	Gerardi et al. (2018); Covas, Rump & Zakrajšek (2014)

Item	Core drivers (form, lag, sign)	Error-correction model	Regime threshold	Source
RWA	Business credit (q/q, L0) +; mortgage credit (q/q, L0) +; consumer credit (q/q, L0) +; unemployment (change, L2) +	Yes: real business and mortgage credit	Unemployment change	FSRA technical briefing (2022); Behn, Haselmann & Wachtel (2016)
Tier 1 capital	Overnight level (L1) +; unemployment (change, L1) −; GDP (q/q, L1) +; 5-year yield (change, L0) −	No (accumulates earnings)	Unemployment change	Borio, Gambacorta & Hofmann (2017); Hirtle et al. (2016); FSRA Q&A (2022)
Tier 2 capital	Disposable income (q/q, L1) +; TSX (q/q, L1) −	No	Income growth	No direct study; share schedule dominates

The system credit and M2 series are proxies. Where your own loan, deposit and income forecasts exist, they are better drivers: interest income from your loan forecasts, interest expense from your deposit forecasts, RWA from your loan mix, and Tier 1 from your net income forecast (the `net_income` concept, left for you to map). The two workbooks that use the overnight rate in levels (margin and Tier 1) keep the metadata file's signs for the data-driven screen, since the screen uses the rate in changes.

All items side by side

Rates help income and capital, and unemployment hurts them; RWA moves with lending. Signs are on each target's quarterly change.

Driver	Interest income	Interest expense	Margin	Non-interest income	Non-interest expense	Provisions	Write-offs	RWA	Tier 1
Overnight rate, change	+	+ (core)	− (core)	0	0	0	0	0	0

Driver	Interest income	Interest expense	Margin	Non-interest income	Non-interest expense	Provisions	Write-offs	RWA	Tier 1
Overnight rate, level	0	0	+ (core)	0	0	0	0	0	+ (core)
Prime rate	+ (core)	+	+	0	0	+ (core, lag 2)	+	0	0
5-year GoC yield	+ (core)	+	+ (core, lag 4)	0	0	0	0	0	− (core, AOCI)
Loan growth	+ (core)	+	0	0	+ (core)	0	0	+ (core)	0
Deposit growth	+	+ (core)	0	0	0	0	0	0	0
Real GDP	0	0	0	+	0	− (core)	−	+	+ (core)
Unemployment	0	0	−	0	0	+ (core)	+ (core)	+ (core)	− (core)
House prices	0	0	0	0	0	−	− (core)	0	+
Equity prices	0	0	0	+ (core)	0	−	0	0	0
Market volatility	0	0	0	− (core)	0	+	0	0	0
Corporate spread	0	0	0	0	0	+ (core)	0	0	−
Wages, CPI	0	0	0	0	+ (core)	0	0	0	0
Insolvencies	0	0	0	0	0	+	+ (core)	+	−

Tier 2 is left out: its only priors are disposable income (+) and equity prices (−), and the share schedule drives it.

Setup checklist before a real run

Most of these items are better driven by your own forecasts than by system proxies, so the order of runs matters.

- [ ] Run balances and rates first, then income, then losses, then capital and RWA, and feed each stage's forecasts into the next as series names.

- [ ] Set the target transform: log growth for interest income and expense, non-interest income and expense, RWA, Tier 1 and Tier 2; change in percentage points for the margin, provision rate and write-off rate.
- [ ] Map `net_income` (Tier 1), `gross_income` (RWA, operational risk) and `shares_redeemable` (Tier 2) to your own series.
- [ ] Check the provision identity in the output: provisions = write-offs + change in the allowance. If the three models disagree, forecast two of them and derive the third.
- [ ] Add a March 2022 break for the FSRA capital rule (RWA, Tier 1/Tier 2 split) and a 2018 break for IFRS 9 (provisions).
- [ ] Confirm the risk weights in Table 2 of the capital rule for each loan class, and split insured from uninsured mortgages for RWA.
- [ ] Compute the capital ratios from the forecasts (Tier 1 ÷ RWA, total ÷ RWA, leverage) and compare with the 6.5%, 8.0%, 10.5% and 3.0% minimums.
- [ ] Run the pipeline's pre-flight check on each workbook.

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Studies cited here and already listed in the earlier reviews (Abad & Suarez; Bikker & Metzemakers; Covas, Rump & Zakrajšek; de Bondt; Drechsler, Savov & Schnabl 2017; Foote, Gerardi & Willen; Gambacorta; Gerardi et al.; Gilchrist & Zakrajšek; Gross & Souleles; Hannan & Berger; Laeven & Majnoni; Neumark & Sharpe; Pugh, Wang & Webley; Sekkel, Takamura & Terajima) carry full citations in the workbooks' references sheets.

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